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Electronic device including barometric pressure sensor

Abstract

An electronic device according to various embodiments of the disclosure includes a front plate configured to form at least a part of a front surface of the electronic device, a speaker disposed in the electronic device, a barometric pressure sensor disposed in the electronic device to measure barometric pressure, a housing configured to form a side surface of the electronic device, and a first slit provided as a gap having a first interval between the housing and an edge portion of the front plate, wherein the housing includes a sound output hole provided at a position at which the speaker is disposed, and a through-hole provided at a position at which the barometric pressure sensor is disposed, and the housing is spaced apart from the front plate to provide a duct configured to connect the through-hole and a sound output passage which is a space for connecting the first slit and the sound output hole. In addition, various other embodiments identified through the specification are possible.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a bypass continuation of International Application No. PCT/KR2022/002586, which was filed on Feb. 22, 2022, and which claims priority to Korean Patent Application No. 10-2021-0023467, filed on Feb. 22, 2021, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entirety.

BACKGROUND

1. Field

(1) Various embodiments of the disclosure relate to an electronic device including a barometric pressure sensor.

2. Description of Related Art

(2) Portable electronic devices are frequently carried during user activities. Recent portable electronic devices may provide users with various functions based on the user activities (for example, hiking, running, and cycling). For example, a portable electronic device may use a barometric pressure sensor mounted therein so as to sense an altitude change in response to the user activity, and may provide information regarding the amount of the user activity (calorie consumption) based thereon. An electronic device having a barometric pressure sensor may include a hole for transferring the barometric pressure of external air to the barometric pressure sensor such that the barometric pressure sensor can measure the barometric pressure.

SUMMARY

(3) A barometric pressure sensor may measure the barometric pressure through efficient inflow of external air. If an electronic device including a barometric pressure sensor includes a hole formed on the outside of the electronic device to transfer the barometric pressure of external air to the barometric pressure sensor, a problem of degraded aesthetic appeal of the electronic device may occur.

(4) Moreover, if a hole for the barometric pressure sensor and an air vent hole through which air can move from inside the electronic device to the outside thereof are used in a shared manner, the waterproof performance of the electronic device may be degraded.

(5) Various embodiments disclosed herein may provide an electronic device including a barometric pressure sensor, which is capable of preventing an external hole of the electronic device from increasing and improving the waterproof performance of the electronic device.

(6) An electronic device according to various embodiments of the disclosure may include a housing including a front plate and a side wall, the front plate is configured to form at least a part of a front surface of the electronic device, and the side wall is configured to form a side surface of the electronic device, a speaker disposed in the housing, a barometric pressure sensor disposed in the housing and configured to measure barometric pressure, and a first slit provided as a gap having a first interval between the side wall and an edge portion of the front plate, wherein the side wall further includes a sound output hole provided at a position at which the speaker is disposed, and a through-hole provided at a position at which the barometric pressure sensor is disposed, and the side wall is spaced apart from the front plate to provide a duct configured to connect the through-hole and a sound output passage which is a space configured to connect the first slit and the sound output hole.

(7) An electronic device according to various embodiments of the disclosure may include a housing including a front plate and a side wall, the front plate is configured to form at least a part of a front surface of the electronic device and the side wall is configured to form a side surface of the electronic device, a speaker disposed in the housing, a barometric pressure sensor disposed in the housing and configured to measure barometric pressure, a first slit provided as a gap having a first interval between the side wall and an edge portion of the front plate, and a first duct configured to connect the first slit and the speaker, wherein the first duct includes a second duct connected to the first duct, and the second duct includes a through-hole provided to allow air introduced from the first slit to be introduced to the barometric pressure sensor.

(8) Based on the above discussion, according to an embodiment of the disclosure, barometric pressure of external air necessary for a barometric pressure sensor may be transferred to the barometric pressure sensor through a speaker hole, and a hole for the barometric pressure sensor may be formed inside the electronic device, thereby reducing the number of holes formed on the outside of the electronic device.

(9) Furthermore, various embodiments of the disclosure may provide an electronic device including a barometric pressure sensor, wherein an air vent hole through which air can move from inside the electronic device to the outside is formed separately from a hole for the barometric pressure sensor, thereby improving the waterproof performance of the electronic device.

(10) Before undertaking the DETAILED DESCRIPTION below, it may be advantageous to set forth definitions of certain words and phrases used throughout this patent document: the terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation; the term “or,” is inclusive, meaning and/or; the phrases “associated with” and “associated therewith,” as well as derivatives thereof, may mean to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, or the like; and the term “controller” means any device, system or part thereof that controls at least one operation, such a device may be implemented in hardware, firmware or software, or some combination of at least two of the same. It should be noted that the functionality associated with any particular controller may be centralized or distributed, whether locally or remotely.

(11) Moreover, various functions described below can be implemented or supported by one or more computer programs, each of which is formed from computer readable program code and embodied in a computer readable medium. The terms “application” and “program” refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer readable program code. The phrase “computer readable program code” includes any type of computer code, including source code, object code, and executable code. The phrase “computer readable medium” includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A “non-transitory” computer readable

medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc or an erasable memory device.

(12) Definitions for certain words and phrases are provided throughout this patent document, those of ordinary skill in the art should understand that in many, if not most instances, such definitions apply to prior, as well as future uses of such defined words and phrases.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of the present disclosure and its advantages, reference is now made to the following description taken in conjunction with the accompanying drawings, in which like reference numerals represent like parts:

(2) FIG. 1 is a perspective view illustrating an electronic device according to one of various embodiments disclosed herein;

(3) FIG. 2 is a perspective view illustrating the electronic device in FIG. 1 viewed from a rear side;

(4) FIG. 3 is a cross-sectional view illustrating the configuration of an electronic device according to an embodiment of the present disclosure;

(5) FIG. 4 is a cross-sectional view illustrating a part of the configuration of an electronic device according to an embodiment of the present disclosure;

(6) FIG. 5 is a conceptual view illustrating the configuration of an electronic device according to an embodiment of the present disclosure;

(7) FIG. 6 is a cross-sectional view illustrating the configuration of an electronic device according to an embodiment of the present disclosure; and

(8) FIG. 7 is a cross-sectional view illustrating the configuration of an electronic device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(9) FIGS. 1 through 7, discussed below, and the various embodiments used to describe the principles of the present disclosure in this patent document are by way of illustration only and should not be construed in any way to limit the scope of the disclosure. Those skilled in the art will understand that the principles of the present disclosure may be implemented in any suitably arranged system or device.

(10) Hereinafter, various embodiments of the disclosure will be described with reference to the accompanying drawings. However, it should be appreciated that they are not intended to limit the disclosure to particular embodiments and include various modifications, equivalents, and/or alternatives for the embodiments of the disclosure. With regard to the description of the drawings, similar reference numerals may be used to designate similar elements.

(11) An electronic device according to various embodiments of the disclosure may include at least one of, for example, a smartphone, a tablet personal computer (PC), a mobile phone, a video phone, an electronic book reader (e-book reader), a desktop PC, a laptop PC, a netbook computer, a workstation, a server, a personal digital assistant (PDA), a portable multimedia player (PMP), a MPEG-1 audio layer-3 (MP3) player, a mobile medical device, a camera, and a wearable device. The wearable device may include at least one of an accessory type (e.g., watch, ring, bracelet, anklet, necklace, glasses, contact lens, or head-mounted device (HMD)), a fabric or clothing-integrated type (e.g., electronic clothing), a body-mounted type (e.g., skin pad, or tattoo), and a bio-implantable type (e.g., implantable circuit).

(12) In some embodiments, the electronic device may include at least one of, for example, a television, a digital video disk (DVD) player, an audio, a refrigerator, an air conditioner, a vacuum

cleaner, an oven, a microwave oven, a washing machine, an air purifier, a set-top box, a home automation control panel, a security control panel, a TV box, a game console, an electronic dictionary, an electronic key, a camcorder, and an electronic photo frame.

(13) In other embodiments, the electronic device may include at least one of various medical devices (e.g., various portable medical measuring devices (blood glucose monitoring device, heart rate monitoring device, blood pressure measuring device, body temperature measuring device, etc.), magnetic resonance angiography (MRA), magnetic resonance imaging (MRI), computed tomography (CT) machine, ultrasonic machine, etc.), a navigation device, a global positioning system (GPS) receiver, an event data recorder (EDR), a flight data recorder (FDR), a vehicle infotainment device, electronic equipment for a ship (e.g., ship navigation device, gyro-compass, etc.), avionics, a security device, an automobile head unit, a home or industrial robot, an automatic teller's machine (ATM) in banks, point of sales (POS) in a shop, or Internet of things devices (e.g., light bulb, various sensors, electric or gas meter, sprinkler device, fire alarm, thermostat, streetlamp, toaster, sporting goods, hot water tank, heater, boiler, etc.).

(14) According to some embodiments, the electronic device may include at least one of a part of furniture or a building/structure, an electronic board, an electronic signature receiving device, a projector, and various kinds of measuring instruments (e.g., water meter, electric meter, gas meter, radio wave meter, etc.). In various embodiments, the electronic device may be one of the aforementioned various devices or a combination of one or more of the aforementioned various devices. In some embodiments, the electronic device may be a flexible electronic device. Further, the electronic device according to embodiments of the disclosure is not limited to the aforementioned devices, and may include a new electronic device according to the development of technology.

(15) FIG. 1 is a perspective view illustrating an electronic device **100** according to various embodiments disclosed herein. FIG. 2 is a perspective view illustrating the electronic device **100** in FIG. 1 viewed from a rear side.

(16) Referring to FIG. 1 and FIG. 2, the electronic device **100** according to an embodiment may include a housing **110** including a first surface (or a front surface) **110A**, a second surface (or a rear surface) **110B**, and a side surface (or a side wall) **110C** surrounding a space between the first surface **110A** and the second surface **110B**. In another embodiment (not illustrated, the term “housing” may refer to a structure forming a part of the first surface **110A**, the second surface **110B**, and the side surface **110C** in FIG. 1.

(17) According to an embodiment, at least a part of the first surface **110A** may be provided by a substantially transparent front plate **102** (e.g., a glass plate or a polymer plate including various coating layers). According to an embodiment, the front plate **102** may include a curved portion extending seamlessly while being curved from the first surface **110A** toward a rear plate **111** in at least one side edge portion.

(18) According to various embodiments, the second surface **110B** may be provided by a substantially opaque rear plate **111**. The rear plate **111** may be made of, for example, coated or colored glass, ceramic, a polymer, or metal (e.g., aluminum, stainless steel (STS), or magnesium), or a combination of two or more of these materials. According to an embodiment, the rear plate **111** may include a curved portion extending seamlessly while being curved from the second surface **110B** towards the front plate **102** in at least one side edge portion.

(19) According to various embodiments, the side surface **110C** may be provided by a side bezel structure (or “a side member or a side wall”) **118** coupled to the front plate **102** and the rear plate **111** and including metal and/or a polymer. In some embodiments, the rear plate **111** and the side bezel structure **118** may be configured in an integral structure, and may include the same material (e.g., a metal material such as aluminum).

(20) According to an embodiment, the electronic device **100** may include at least one of a display **101**, an audio module **103** and **114a**, a sensor module, a camera module **105**, a key input device

117, and a connector hole **108**. In some embodiments, in the electronic device **100**, at least one of the components (e.g., the key input devices **117**) may be omitted, or other components may be additionally included. For example, the electronic device **100** may further include a sensor module (not illustrated). For example, in an area provided by the front plate **102**, a sensor such as a proximity sensor or an illuminance sensor may be integrated with the display **101** or may be disposed at a position adjacent to the display **101**. In some embodiments, the electronic device **100** may further include a light-emitting element, and the light-emitting element may be disposed at a position adjacent to the display **101** in the area provided by the front plate **102**. For example, the light-emitting element may provide the status information of the electronic device **100** in an optical form. In another example, the light-emitting element may provide a light source which is interlocked with the operation of the camera module **105**. The light-emitting element may include, for example, an LED, an IR LED, and a xenon lamp.

(21) For example, the display **101** may be exposed through a substantial portion of the front plate **102**. In some embodiments, the edge of the display **101** may be configured to have substantially the same shape as the shape of the outline (e.g., a curved surface) of the front plate **102** adjacent thereto. In another embodiment (not illustrated), the interval between the outline of the display **101** and the outline of the front plate **102** may be substantially constant in order to enlarge the area in which the display **101** is exposed. In another embodiment (not illustrated), the electronic device may have a recess or an opening provided in a portion of the screen display area of the display **101** and include another electronic component, such as the camera module **105**, a proximity sensor (not illustrated), or an illuminance sensor (not illustrated) aligned with the recess or the opening provided.

(22) In another embodiment (not illustrated), the rear surface of the screen display area of the display **101** may include at least one of the camera modules **112** and **113**, a fingerprint sensor **116**, and a flash **106**. In another embodiment (not illustrated), the display **101** may be coupled to or disposed adjacent to a touch-sensitive circuit, a pressure sensor capable of measuring a touch intensity (pressure), and/or a digitizer configured to detect a magnetic field-type stylus pen.

(23) The audio module **103** and **114a** may include a microphone hole and a speaker hole. A microphone may be disposed in the microphone hole to acquire external sound, and in some embodiments, multiple microphones may be disposed therein to detect the direction of sound. In some embodiments, the speaker hole and the microphone hole may be implemented as a single hole **103**, or a speaker (e.g., a piezo speaker) may be included without a speaker hole. The speaker hole may include an external speaker hole and a call receiver hole **114a**.

(24) The electronic device **100** may generate an electrical signal or a data value corresponding to an internal operating state or an external environmental state by including a sensor module (not illustrated). The sensor module may further include, for example, a proximity sensor disposed on the first surface **110A** of the housing **110**, a fingerprint sensor integrated with or disposed adjacent to the display **101**, and/or a biometric sensor (e.g., an HRM sensor) disposed on the second surface **110B** of the housing **110**. The electronic device **100** may further include at least one of sensor modules (not illustrated), such as a gesture sensor, a gyro sensor, a barometric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(25) The camera modules **105**, **112**, **113**, and **106** may include a first camera device **105** disposed on the first surface **110A** of the electronic device **100**, second camera devices **112** and **113** disposed on the second surface **110B** thereof, and/or a flash **106**. The camera modules **105**, **112**, and **113** may include one or more lenses, an image sensor, and/or an image signal processor. The flash **106** may include, for example, a light-emitting diode or a xenon lamp. In some embodiments, two or more lenses (an infrared camera lens, a wide-angle lens, and a telephoto lens) and image sensors may be disposed on one surface of the electronic device **100**.

(26) The key input devices **117** may be disposed on the side surface **110C** of the housing **110**. In

another embodiment, the electronic device **100** may not include some or all of the above-mentioned key input devices **117**, and a key input device **117**, which is not included in the above-mentioned key input devices, may be implemented in another form, such as a soft key, on the display **101**. In some embodiments, the key input devices may include at least a part of a fingerprint sensor **116** disposed on the second surface **110B** of the housing **110**.

(27) The connector hole **108** may accommodate a connector configured to transmit and receive power and/or data to and from an external electronic device, and/or a connector configured to transmit and receive an audio signal to and from an external electronic device. For example, the connector hole **108** may include a USB connector or an earphone jack.

(28) FIG. **3** is a cross-sectional view illustrating an internal configuration of an electronic device according to an embodiment.

(29) Referring to FIG. **1** to FIG. **3**, an electronic device **100** according to an embodiment may include a housing **110** including a first surface (or a front surface) **110A**, a second surface (or a rear surface) **110B**, and a side surface (or a side wall) **110C** surrounding a space between the first surface **110A** and the second surface **110B**. In another embodiment (not illustrated, the term “housing” may refer to a structure forming a part of the first surface **110A**, the second surface **110B**, and the side surface **110C** in FIG. **1**. According to an embodiment, the electronic device **100** may include a front plate (front **102** in FIG. **1**) forming at least a part of the front surface **110A** of the electronic device **100**.

(30) Referring to FIG. **3**, the electronic device **100** according to an embodiment may include at least one opening opened to the outside of the electronic device **100**. According to an embodiment, the at least one opening may include a first slit **114**. The first slit **114** may be provided as a gap having an interval between the side wall **110C** of the housing **110** and an edge portion of the front plate (front plate **102** in FIG. **1**). However, the shape of the first slit **114** is not limited thereto. For example, the first slit **114** may have a shape of a hole provided in the front plate (front plate **102** in FIG. **1**). The first slit **114** may have a shape of a hole provided on the side surface **110C**. According to another embodiment, the first slit **114** may have a shape of an air vent hole. According to an embodiment, the first slit **114** may include a call receiver hole **114a** or a speaker hole.

(31) According to an embodiment, the electronic device **100** may further include at least one second slit **115**. The at least one second slit **115** may be provided as a gap having an interval between the side wall **110C** of the housing **110** and an edge of the front plate (front plate **102** in FIG. **1**). The interval of the second slit **115** may be provided to be smaller than the interval of the first slit **114**. For example, the interval of the first slit **114** may be provided to be an interval of about 0.3 mm. The second slit **115** may be a gap of about 0.05 mm provided along the edge portion of the front plate (front plate **102** in FIG. **1**) between the housing **110** and the front plate. However, the shape of the second slit is not limited thereto. For example, the second slit **115** may have a shape of a hole provided in the front plate (front plate **102** in FIG. **1**). The second slit **115** may have a shape of a hole provided on the side surface **110C**. According to another embodiment, the second slit **115** may have a shape of an air vent hole.

(32) According to an embodiment, the first slit **114** may be provided at an upper end of the electronic device **100**. According to an embodiment, the second slit **115** may be provided at an upper end of the electronic device **100**. However, those are not limited thereto. For example, the first slit **114** and the second slit **115** may be provided at a lower end or a side portion of the electronic device **100**. According to an embodiment, the second slit **115** may be provided to extend from the first slit **114**.

(33) According to an embodiment, the electronic device **100** may include a speaker **130** and a barometric pressure sensor **119**. The speaker **130** may be disposed in the electronic device. According to an embodiment, the speaker **130** may be disposed to correspond to a position at which the first slit **114** is provided. According to an embodiment, the speaker **130** may be disposed to be adjacent to the first slit **114**. According to an embodiment, the barometric pressure sensor **119** may be disposed in the electronic device **100**. According to an embodiment, the barometric

pressure sensor **119** may be disposed to be adjacent to the first slit **114**. According to an embodiment, the barometric pressure sensor **119** may be a waterproof barometric pressure sensor.

(34) According to an embodiment, the electronic device **100** may include a duct **200**. According to an embodiment, the duct **200** may include at least a part of a space provided between the front plate **102** and the side wall **110C** of the housing **110**. According to an embodiment, the duct **200** may be a space defined by a waterproof member **120** disposed between the front plate **102** and the side wall **110C** of the housing **110** in the electronic device **100**. According to another embodiment, the duct **200** may be provided to have a shape of a separate passage. According to an embodiment, the duct **200** may include at least one of a first duct **200A**, a second duct **200B**, or a third duct **200C**. According to an embodiment, the first duct **200A** may connect the first slit **114** and the speaker **130**. According to an embodiment, the first duct **200A** may include a sound output hole. The sound output hole may be provided to correspond to a position at which the speaker **130** is disposed. The sound output hole may be provided at the housing **110**. According to an embodiment, the first duct **200A** may include the second duct **200B**. According to an embodiment, the second duct **200B** may be provided to extend from at least one surface of the first duct **200A**. According to an embodiment, the third duct **200C** may be provided to extend from at least one surface of the first duct **200A**. According to an embodiment, the second duct **200B** may be provided along the edge portion of the housing **110**. At least one of the first duct **200A**, the second duct **200B**, or the third duct **200C** may be provided as a space surrounded by the waterproof member **120** configured to bond the housing **110**, the front plate (front plate **102** in FIG. 1), and a space between the housing **110** and the front plate (front plate **102** in FIG. 1). According to an embodiment, the second duct **200B** may be provided to extend from the first duct **200A** in an x-axis direction. According to an embodiment, the second duct **200B** may be provided at the upper end of the electronic device **100**. According to an embodiment, the second duct **200B** may connect the first duct **200A** and the barometric pressure sensor **119**. Accordingly, the barometric pressure sensor **119** may be connected to the outside via the first slit **114**. As the barometric pressure sensor **119** is connected to the outside of the electronic device **100** via the passage through which the sound output from the speaker **130** is output, a separate hole for a barometric pressure sensor may not be provided at the outside of the electronic device **100**.

(35) According to an embodiment, the third duct **200C** may be provided to extend from the second duct **200B**. According to an embodiment, the third duct **200C** may be provided to extend from the first duct **200A**. However, the shapes of the first duct **200A**, the second duct **200B**, and the third duct **200C** are not limited thereto. For example, the first duct **200A**, the second duct **200B**, and the third duct **200C** may be provided as a single duct **200**. According to an embodiment, the third duct **200C** may be provided along the edge portion of the housing **110**. According to an embodiment, the third duct **200C** may be provided to extend along a side edge of the housing **110** in -y-axis direction.

(36) According to an embodiment, the first slit **114** may be connected with the first duct **200A**. The first slit **114** may be directly connected to the second duct **200B** or the third duct **200C**. According to an embodiment, the second slit **115** may be directly connected to at least one of the first duct **200A**, the second duct **200B**, and the third duct **200C**.

(37) According to an embodiment, the electronic device **100** may include an inner hole connected with the barometric pressure sensor **119**. The inner hole may be provided to have a shape of a through-hole **202**. According to an embodiment, the through-hole **202** may include at least a part of a space provided between the housing **110** and the front plate **102**. According to another embodiment, the through-hole **202** may be provided at the housing **110**. The through-hole **202** may be provided at a position at which the barometric pressure sensor **119** is disposed. According to an embodiment, the through-hole **202** may be spaced apart from the first slit **114**. According to an embodiment, the through-hole **202** may be spaced apart from the first slit **114** in the x-axis direction.

(38) The electronic device **100** may further include an air vent hole through which air can move from the inside to the outside of the electronic device **100** at the outside of the electronic device **100**. An electronic device (not illustrated) according to a comparative embodiment may include a non-waterproof barometric pressure sensor. When the electronic device includes a non-waterproof barometric pressure sensor, a hole configured to allow barometric pressure of external air to be transmitted to the barometric pressure sensor and an air vent hole configured to allow air to move from the inside to the outside of the housing **110** may be provided as a single hole so that the barometric pressure sensor can measure the barometric pressure. However, the electronic device including the non-waterproof barometric pressure sensor may include a waterproof film for the non-waterproof barometric pressure sensor. When the waterproof film is applied, there may be a limitation in waterproof performance of the electronic device. On the other hand, the electronic device **100** according to an embodiment of the disclosure may include a waterproof barometric pressure sensor. The electronic device **100** including the waterproof barometric pressure sensor may not include the waterproof film for the barometric pressure sensor. According to an embodiment, the electronic device **100** may have a separate air vent hole (not illustrated) from the first slit **114** and the second slit **115**, which is provided at the outside of the electronic device **100**. According to an embodiment, the air vent hole may be provided at the housing **110**.

(39) According to an embodiment, the electronic device **100** may further include the waterproof member **120**. According to an embodiment, the waterproof member **120** may be disposed to be spaced apart from the edge portion of the housing **110**. According to an embodiment, the duct **200** and the through-hole **202** may be provided between the waterproof member **120** and the edge portion of the housing **110**. The waterproof member **120** may block foreign substances such as water or dust introduced into the housing **110**.

(40) FIG. **4** is a cross-sectional view illustrating a part of the configuration of an electronic device according to an embodiment of the present disclosure.

(41) The configuration of the electronic device **100** may be referred to by components of the electronic device **100** in FIG. **3**. The same numerals are used for the same or substantially the same components as the contents described above, and a redundant description will be omitted.

(42) Referring to FIGS. **3** and **4**, the electronic device **100** according to an embodiment may accommodate external air. According to an embodiment, the electronic device **100** may accommodate the external air via the first slit **114**. According to an embodiment, the external air may be introduced in a direction **300** from the outside of the electronic device **100** toward the inside of the electronic device **100** via the first slit **114**. A narrow gap through which air can pass may be provided between the side surface **110C** portion of the housing and the front plate (front plate **102** in FIG. **1**) in addition to the first slit **114**. The gap provided between the side surface **110C** portion of the housing and the front plate (front plate **102** in FIG. **1**) in addition to the first slit **114** may be referred to as the second slit **115**. According to an embodiment, the electronic device **100** may accommodate the external air via the second slit **115**.

(43) According to an embodiment, air introduced from the first slit **114** and/or the second slit **115** may move along the duct **200**. According to an embodiment, the introduced air may move along the first duct **200A**. The air moved along the first duct **200A** may move along the second duct **200B**. According to an embodiment, the air moved along the second duct **200B** may move along the third duct **200C**. The air moved along the first duct **200A** may move along the third duct **200C**. Air may be introduced into the through-hole **202** along the second duct **200B**. According to an embodiment, the barometric pressure sensor **119** may be configured to acquire external information based on the air introduced through the through-hole **202**. According to an embodiment, the electronic device **100** may include a display (display **101** in FIG. **1**) or an audio module (audio module **103** and receiver hole **114a** in FIG. **1**). The processor may be configured to output contents related to the information via the display (display **101** in FIG. **1**) or the audio module (audio module **103** and receiver hole **114a** in FIG. **1**) based on the information acquired by the barometric

pressure sensor **119**.

(44) According to an embodiment, the electronic device **100** may further include the waterproof member **120** disposed within the housing **110**. The waterproof member **120** may seal the housing **110** to prevent foreign substances such as water or dust from being introduced from the outside of the housing **110** into the housing **110**. According to an embodiment, the waterproof member **120** may be disposed to be spaced apart from the edge portion of the housing **110**. According to an embodiment, the waterproof member **120** may be disposed to be spaced apart from the edge portion of the side surface **110C**. According to an embodiment, the duct **200** and the through-hole **202** may be provided between the edge portion of the side surface **110C** and the waterproof member **120**. As the waterproof member **120** is disposed to be spaced apart from the side surface **110C**, the duct **200** may be provided along the edge portion of the side surface **110C**.

(45) FIG. 5 is a conceptual view illustrating the configuration of an electronic device according to an embodiment of the present disclosure.

(46) Referring to FIG. 5, the electronic device **100** may include the housing **110**, the speaker **130**, the barometric pressure sensor **119**, and the waterproof member **120**. According to an embodiment, the speaker **130**, the barometric pressure sensor **119**, and the waterproof member **120** may be arranged in the electronic device **100**. According to an embodiment, the electronic device **100** may include the first slit **114**. According to an embodiment, the electronic device **100** may further include at least one second slit **115**. According to an embodiment, the first slit **114** and the at least one second slit **115** may be opened to the outside of the electronic device **100**. According to an embodiment, external air may be introduced from the first slit **114** and the second slit **115** into the electronic device **100**.

(47) According to an embodiment, the first slit **114** and the second slit **115** may be provided to an outside the electronic device **100**. According to an embodiment, the first slit **114** and the second slit **115** may be provided at the housing **110**. According to an embodiment, the first slit **114** may include a speaker hole or a call receiver hole **114a**.

(48) According to an embodiment, the electronic device **100** may include the first duct **200A**, the second duct **200B**, the third duct **200C**, and the through-hole **202**. The first duct **200A**, the second duct **200B**, the third duct **200C**, and the through-hole **202** may be provided in the electronic device **100**. However, the configuration of the electronic device **100** is not limited thereto. For example, the electronic device **100** may exclude at least one of the components described above, or may further include at least one of other components.

(49) According to an embodiment, the first duct **200A**, the second duct **200B**, and the third duct **200C** may include at least a part of the space provided between the housing **110** and the front plate (front plate **102** in FIG. 1). According to an embodiment, the first duct **200A**, the second duct **200B**, and the third duct **200C** may be a space defined by the waterproof member **120** disposed between the housing **110** and the front plate **102** in the electronic device. The first duct **200A**, the second duct **200B**, and the third duct **200C** may have separate passage shapes and may be provided at the housing **110**.

(50) According to an embodiment, the first duct **200A** may include the second duct **200B**. According to an embodiment, the second duct **200B** may be provided to extend from the first duct **200A**. According to an embodiment, the third duct **200C** may be provided to extend from the second duct **200B**. According to an embodiment, the first duct **200A** may include the third duct **200C**. According to an embodiment, the third duct **200C** may be provided to extend from the first duct **200A**. The third duct **200C** provided to extend from the second duct **200B** may be connected to the first duct **200A** along the edge portion of the housing **110**. According to an embodiment, the first duct **200A**, the second duct **200B**, and the third duct **200C** may be provided as one duct **200**.

(51) According to an embodiment, the second duct **200B** may include the through-hole **202** communicating with the barometric pressure sensor **119**. According to an embodiment, the through-hole **202** may be spaced apart from the first slit **114**. According to an embodiment, the

barometric pressure sensor **119** may be disposed adjacent to the through-hole **202**. According to an embodiment, the barometric pressure sensor **119** may be disposed adjacent to the speaker **130**.

(52) According to an embodiment, the first slit **114** may be connected to the first duct **200A**. The first duct **200A** may connect the first slit **114** and the speaker **130**. According to an embodiment, the second slit **115** may be connected with the second duct **200B**. According to an embodiment, the second slit **115** may be connected with the first duct **200A** and the third duct **200C**. According to an embodiment, the first duct **200A**, the second duct **200B**, the third duct **200C**, and the through-hole **202** may be arranged between the housing **110** and the waterproof member **120**. According to an embodiment, the waterproof member **120** may be provided to be disposed along the first duct **200A**, the second duct **200B**, the third duct **200C**, and the through-hole **202**.

(53) According to an embodiment, the electronic device **100** may further include a processor. The barometric pressure sensor **119** may be electrically connected with the processor. The barometric pressure sensor **119** may be configured to acquire external information based on the air introduced through the first slit **114** and the at least one second slit **115**. According to an embodiment, the electronic device **100** may include a display (display **101** in FIG. **1**) or an audio module (audio module **103** and receiver hole **114a** in FIG. **1**). The processor may be configured to output contents related to the information through the display (display **101** in FIG. **1**) or the audio module (audio module **103** and receiver hole **114a** in FIG. **1**) based on the information acquired by the barometric pressure sensor **119**.

(54) FIG. **6** is a cross-sectional view illustrating the configuration of an electronic device according to an embodiment of the present disclosure.

(55) Referring to FIG. **6**, external air introduced from the first slit **114** into the electronic device **100** may pass through the through-hole **202**. The air passed through the through-hole **202** may move along the second duct **200B** in $-y$ -axis direction. The barometric pressure sensor **119** may acquire air moving along the second duct **200B**.

(56) According to an embodiment, the electronic device **100** may include a printed circuit board (PCB) and a main board (not illustrated). According to an embodiment, the printed circuit board (PCB) and the main board (not illustrated) may be arranged in the electronic device **100**. According to an embodiment, the printed circuit board (PCB) may be a flexible PCB (FPCB) **122**. According to an embodiment, as the barometric pressure sensor **119** is disposed close to the first slit **114**, the electronic device **100** may include the FPCB **122** to connect the barometric pressure sensor **119** and the main board (not illustrated). According to an embodiment, the FPCB **122** may be disposed adjacent to the barometric pressure sensor **119**. According to an embodiment, the FPCB **122** may face the barometric pressure sensor **119**. According to an embodiment, the FPCB **122** may be electrically connected with the barometric pressure sensor **119**. The FPCB **122** may be electrically connected with the main board (not illustrated).

(57) According to an embodiment, the electronic device **100** may further include a waterproof member **121**. The waterproof member **121** may be disposed between the housing **110** and the barometric pressure sensor **119**. The waterproof member **121** may be disposed between the second duct **200B** and the barometric pressure sensor **119**. The waterproof member **121** may block foreign substances such as water or dust introduced into a space between the housing **110** or the second duct **200B** and the barometric pressure sensor **119**. According to an embodiment, the waterproof member **121** may include a shape of an O-ring. However, the shape of the waterproof member **121** is not limited thereto. According to an embodiment, the waterproof member **121** may include a material for sealing a gap between the barometric pressure sensor **119** and the housing **110C** to block foreign substances such as water or dust. For example, the waterproof member **121** may include an O-ring made of a rubber or silicone material. However, the material of the waterproof member **121** is not limited thereto. For example, the waterproof member **121** may be made of a material for allowing inflow/outflow of air but restricting inflow/outflow of water or moisture. According to another embodiment, the waterproof member **121** may be made of a material for

restricting not only inflow/outflow of air but also inflow/outflow water or humidity.

(58) FIG. 7 is a cross-sectional view illustrating the configuration of an electronic device according to an embodiment of the present disclosure.

(59) The configuration of the electronic device **100** may be referred to by components of the electronic device **100** of FIG. 1 to FIG. 7. The same numerals are used for the same or substantially the same components as the contents described above, and a redundant description will be omitted.

(60) Referring to FIG. 7, at least a part of the duct **200** may include an inclined surface. According to an embodiment, at least a part of the second duct **200B** configured to connect the through-hole **202** and the barometric pressure sensor **119** may include an inclined surface. The inclined surface may be provided to form an inclination angle with one surface of the front plate **102**. The inclined surface may include at least a part of a space provided between the front plate **102** and the housing **110**. According to an embodiment, the inclined surface may be a space defined by the waterproof member **120** disposed between the front plate **102** and the housing **110** in the electronic device, but is not limited thereto. For example, the duct including the inclined surface may be provided as a separate passage.

(61) According to an embodiment, water may be introduced from the outside of the electronic device **100** into the electronic device **100**. According to an embodiment, water may be introduced to the first slit **114** and/or the second slit **115**. The introduced water may move along the duct **200**. The inclined surface according to an embodiment may help water moving along the duct **200** to be discharged to the outside of the electronic device **100**.

(62) The various embodiments of the disclosure disclosed in the specification and drawings are merely provided as specific examples to easily explain the technical content of the disclosure and help the understanding of the disclosure, and are not intended to limit the scope of the disclosure. Therefore, the scope of the disclosure should be construed that all changes or modified forms derived based on the technical idea of the disclosure in addition to the embodiments disclosed herein are included in the scope of the disclosure.

(63) As described above, an electronic device (e.g., the electronic device **100** in FIG. 1) according to an embodiment may include a housing with a front plate and a side wall, the front plate can be configured to form at least a part of a front surface of the electronic device, and a side wall can be configured to form a side surface of the electronic device, a speaker disposed in the housing, a barometric pressure sensor disposed in the housing and configured to measure barometric pressure, and a first slit provided as a gap having a first interval between the side wall and an edge portion of the front plate, wherein the housing includes a sound output hole provided at a position at which the speaker is disposed and a through-hole provided at a position at which the barometric pressure sensor is disposed, and wherein the side wall is spaced apart from the front plate to provide a duct configured to connect the through-hole and a sound output passage which is a space for connecting the first slit and the sound output hole.

(64) According to an embodiment, the electronic device may further include a second slit provided as at least one gap having a second interval between the side wall and an edge portion of the front plate, and the second slit may be connected to the duct.

(65) According to an embodiment, the electronic device may further include a waterproof member disposed in the housing, the waterproof member may be disposed to be spaced apart from the edge portion of the side wall, and the duct and the through-hole may be provided between the edge portion of the side wall and the waterproof member.

(66) According to an embodiment, the electronic device may further include a flexible PCB (FPCB) that may face the barometric pressure sensor.

(67) According to an embodiment, the electronic device may further include a main board that may be electrically connected with the flexible PCB (FPCB).

(68) According to an embodiment, the side wall may be spaced apart from the front plate to further provide another duct connected with the duct along the edge portion of the front plate.

(69) According to an embodiment, the side wall may further include an air vent hole through which air can move from the inside to the outside of the housing.

(70) According to an embodiment, the through-hole may be spaced apart from the first slit.

(71) According to an embodiment, at least a part of the duct may include an inclined surface disposed to form an inclination angle with a facing surface of the front plate.

(72) According to an embodiment, the electronic device may further include a waterproof member disposed between the barometric pressure sensor and the through-hole.

(73) As described above, an electronic device (e.g., the electronic device **100** in FIG. **1**) according to an embodiment may include a housing with a front plate and a side wall, the front plate can be configured to form at least a part of a front surface of the electronic device, and the side wall can be configured to form at least a part of a side surface of the electronic device, a speaker disposed in the housing, a barometric pressure sensor disposed in the housing and configured to measure barometric pressure, a first slit provided as a gap having a first interval between the side wall and an edge portion of the front plate, and a first duct configured to connect the first slit and the speaker, wherein the first duct includes a second duct connected to the first duct, and the second duct includes a through-hole provided to allow air introduced from the first slit to be introduced to the barometric pressure sensor.

(74) According to an embodiment, the electronic device may further include a second slit provided as at least one gap having a second interval between the side wall and an edge of the front plate, and the second slit may be connected to the second duct.

(75) According to an embodiment, the electronic device may further include a waterproof member disposed in the housing, the waterproof member may be disposed to be spaced apart from the edge portion of the side wall, and the second duct and the through-hole may be provided between the edge portion of the side wall and the waterproof member.

(76) According to an embodiment, the electronic device may further include a flexible PCB (FPCB), that may face the barometric pressure sensor.

(77) According to an embodiment, the electronic device may further include a main board that may be electrically connected with the flexible PCB (FPCB).

(78) According to an embodiment, the electronic device may further include a third duct provided along the edge portion of the front plate, and the third duct may be connected with the second duct.

(79) According to an embodiment, the side wall may further include an air vent hole through which air can move from the inside to the outside of the housing.

(80) According to an embodiment, the through-hole may be spaced apart from the first slit.

(81) According to an embodiment, at least a part of the second duct may include an inclined surface disposed to form an inclination angle with a facing surface of the front plate.

(82) According to an embodiment, the electronic device may further include a waterproof member disposed between the barometric pressure sensor and the second duct.

(83) Although the present disclosure has been described with various embodiments, various changes and modifications may be suggested to one skilled in the art. It is intended that the present disclosure encompass such changes and modifications as fall within the scope of the appended claims.

Claims

1. An electronic device comprising: a housing including: a front plate configured to form at least a part of a front surface of the electronic device, and a side wall configured to form a side surface of the electronic device; a speaker disposed in the housing; a barometric pressure sensor disposed in the housing and configured to measure barometric pressure; and a first slit provided as a gap having a first interval from the side wall to an edge portion of the front plate; and a second slit provided as at least one gap having a second interval between the side wall and an edge portion of the front

- plate, wherein the side wall further includes: a sound output hole provided at a position at which the speaker is disposed, and a through-hole provided at a position at which the barometric pressure sensor is disposed, wherein the side wall is spaced apart from the front plate to provide a duct configured to connect the through-hole and a sound output passage which is a space configured to connect the first slit and the sound output hole, and wherein the duct includes: a first duct configured to connect the first slit and the speaker, and a second duct that extends from the first duct and configured to connect the second slit and the barometric pressure sensor.
2. The electronic device of claim 1, wherein the second interval defines a different distance than the first interval.
 3. The electronic device of claim 1, further comprising a waterproof member disposed in the housing, wherein the waterproof member is disposed to be spaced apart from an edge portion of the side wall, and wherein the duct and the through-hole are provided between the edge portion of the side wall and the waterproof member.
 4. The electronic device of claim 1, further comprising a flexible PCB (FPCB) that faces the barometric pressure sensor.
 5. The electronic device of claim 4, further comprising a main board electrically connected to the flexible PCB (FPCB).
 6. The electronic device of claim 1, wherein the side wall is spaced apart from the front plate to further provide another duct connected with the duct along the edge portion of the front plate.
 7. The electronic device of claim 1, wherein the side wall further includes an air vent hole through which air can move from an inside to an outside of the housing.
 8. The electronic device of claim 1, wherein the through-hole is spaced apart from the first slit.
 9. The electronic device of claim 1, wherein at least a part of the duct includes an inclined surface disposed to form an inclination angle with a facing surface of the front plate.
 10. The electronic device of claim 1, further comprising a waterproof member disposed between the through-hole and the barometric pressure sensor.
 11. An electronic device comprising: a housing including: a front plate configured to form at least a part of a front surface of the electronic device, and a side wall configured to form a side surface of the electronic device; a speaker disposed in the housing; a barometric pressure sensor disposed in the housing and configured to measure barometric pressure; a first slit provided as a gap having a first interval from the side wall to an edge portion of the front plate; a second slit provided as at least one gap having a second interval between the side wall and an edge portion of the front plate; and a first duct configured to connect the first slit and the speaker, wherein the first duct includes a second duct connected to and extending from the first duct and configured to connect the second slit and the barometric pressure sensor, and wherein the second duct includes a through-hole provided to allow air introduced from the first slit to be introduced to the barometric pressure sensor.
 12. The electronic device of claim 11, wherein the second interval defines a different distance than the first interval.
 13. The electronic device of claim 11, further comprising a waterproof member disposed in the housing, wherein the waterproof member is spaced apart from an edge portion of the side wall, and wherein the second duct and the through-hole are provided between the edge portion of the side wall and the waterproof member.
 14. The electronic device of claim 11, further comprising a flexible PCB (FPCB) that faces the barometric pressure sensor.
 15. The electronic device of claim 14, further comprising a main board electrically connected to the flexible PCB (FPCB).
 16. The electronic device of claim 11, further comprising a third duct provided along the edge portion of the front plate, wherein the third duct is connected with the second duct.
 17. The electronic device of claim 11, wherein the side wall further includes an air vent hole

through which air can move from an inside to an outside of the housing.

18. The electronic device of claim 11, wherein the through-hole is spaced apart from the first slit.

19. The electronic device of claim 11, wherein at least a part of the second duct includes an inclined surface disposed to form an inclination angle with a facing surface of the front plate.

20. The electronic device of claim 11, further comprising a waterproof member disposed between the second duct and the barometric pressure sensor.
